

Market Basket Analysis for Retail Product Recommendation Using Association Rule Mining with RFM-Based Customer Segmentation

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Abstract: This paper presents a Market Basket Analysis (MBA) system for retail product recommendation using Association Rule Mining, specifically the Apriori Algorithm. The system was evaluated on the UCI Online Retail II dataset containing over one million real UK retail transactions from December 2009 to December 2011. Four key contributions beyond standard Apriori are introduced: (1) conviction-based rule pruning, (2) RFM-based customer segmentation with per-segment rule mining, (3) temporal purchase pattern analysis, and (4) algorithmic scalability measurement. The system generated 461 high-quality association rules with an average lift of 10.57, classified as exceptional. A personalized recommendation engine was deployed that delivers segment-specific product suggestions based on customer behavioral tiers.

Index Terms. Apriori algorithm, association rule mining, customer segmentation, data mining, market basket analysis, retail recommendation system.

I. INTRODUCTION

Retail businesses generate enormous volumes of transaction data daily. Hidden within these records are co-purchase patterns that Market Basket Analysis (MBA) can uncover automatically. The Apriori Algorithm, introduced by Agrawal and Srikant [1], remains the foundational method for association rule mining. It operates by scanning transaction data in multiple passes to identify frequently co-purchased item combinations and converts them into actionable IF-THEN rules.

A review of published literature from 2020 to 2025 identified four significant gaps in existing Apriori applications: (i) all customers are treated identically; (ii) the temporal dimension of transactions is ignored; (iii) only support and confidence are used as filters, producing large numbers of weak redundant rules; and (iv) algorithmic scalability is never measured. This paper addresses all four gaps and demonstrates that targeted enhancements substantially improve the quality and personalization of generated recommendations.

II. RELATED WORK

Chen et al. [2] applied RFM scoring to online retail data for customer segmentation but did not integrate segmentation into the association rule mining pipeline. Hoque et al. [3] proposed customer-centric MBA using consumer segmentation and demonstrated improved rule robustness; however, scalability measurement and temporal analysis were not addressed. Vaishampayan et al. [4] studied efficient MBA using association rule mining but used only support and confidence as filters. Kannan and Immanuel [6] improved the Apriori candidate generation phase but did not consider per-segment rule mining. The present work synthesizes these contributions by combining RFM segmentation, conviction-based pruning, temporal analysis, and scalability measurement into a unified pipeline.

III. DATASET

A. Dataset Description

The UCI Online Retail II dataset [5] was selected after evaluating multiple alternatives. The dataset originates from a UK-based gift and home decor retailer and contains real, non-synthetic transactional data. Table I summarizes the key dataset attributes.

TABLE I. DATASET SUMMARY

Attribute	Value
Source	UCI Online Retail II
Period	Dec 2009 to Dec 2011
Total Rows	1,067,371 records
After Cleaning	31,724 unique invoices
Unique Customers	5,878
Products Used	200 (top by frequency)
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B. Data Preprocessing

Raw data preprocessing involved: (i) removal of cancelled invoices starting with C; (ii) exclusion of records with missing Customer IDs; (iii) removal of negative-quantity returns and zero-price records; (iv) computation of TotalPrice as Quantity multiplied by Price for RFM monetary scoring. After cleaning, 31,724 valid invoices were retained.

IV. METHODOLOGY

A. RFM Customer Segmentation

Every customer was scored across three dimensions using their full purchase history. Recency measures days since the most recent purchase. Frequency counts unique invoices placed. Monetary records total spend. Each dimension was divided into three equal quantiles and scored 1 to 3. The composite RFM Score ranges from 3 to 9. Customers scoring 8 or 9 were classified as High Value, scores 5 to 7 as Medium Value, and scores 3 or 4 as Low Value.

B. Temporal Analysis

The InvoiceDate column was decomposed into month, day-of-week, hour-of-day, and quarter. Monthly revenue and transaction volume trends were plotted to identify seasonal peaks. Day-of-week and hourly distributions revealed high-activity purchasing windows, directly filling the temporal gap identified in the literature review.

C. Apriori Algorithm on Full Dataset

The dataset was reduced to the top 200 products by frequency to ensure memory feasibility. A binary transaction matrix was constructed where rows represent invoices and columns represent products. Table II presents the Apriori parameter configuration used in this study.

TABLE II. APRIORI ALGORITHM PARAMETERS

Parameter	Value	Rationale
Min. Support	0.01	Item must appear in at least 1% of baskets
Min. Confidence	0.20	Rule must be correct at least 20% of the time
Min. Lift	1.0	Rule must exceed random chance
Min. Conviction	1.1	Removes accidental and redundant rules
Max. Rule Length	3	Single items, pairs, and triplets only

D. Per-Segment Apriori

The complete Apriori process was repeated independently for each of the three RFM customer segments. High, Medium, and Low Value customers each received separate frequent itemset mining and rule generation. A High Value customer receives recommendations derived exclusively from the purchasing behavior of other High Value customers. This is the core novel contribution of the project.

V. RESULTS AND EVALUATION

A. Rule Quality KPIs

Table III reports the key performance indicators computed across all 461 generated association rules.

TABLE III. RULE QUALITY KEY PERFORMANCE INDICATORS

KPI	Definition	Result
Support	$\text{freq}(A \text{ and } B) \text{ divided by total transactions}$	Average 0.0141 across 461 rules
Confidence	$\text{freq}(A \text{ and } B) \text{ divided by } \text{freq}(A)$	Average 40.2%
Lift	Confidence divided by $P(B)$	Average 10.57, classified as exceptional
Conviction	$(1 \text{ minus } P(B)) \text{ divided by } (1 \text{ minus Confidence})$	Average 1.75
Total Rules	Valid rules after all filters applied	461 from 444 frequent itemsets

B. Algorithm Scalability

Table IV reports execution time and rule counts per segment, demonstrating that the per-segment approach scales efficiently from 13.64 seconds on the full dataset to 0.08 seconds on the Low Value segment.

TABLE IV. ALGORITHM SCALABILITY METRICS

Segment	Customers	Transactions	Rules	Avg Lift	Time (s)
Full Dataset	5,878	31,724	461	10.57	13.64
High Value	1,734	22,812	75	8.45	3.83
Medium Value	2,332	7,168	29	10.75	0.62
Low Value	1,461	1,744	10	9.13	0.08

C. Top Association Rules

Table V lists the five highest-lift rules generated from the full dataset.

TABLE V. TOP 5 ASSOCIATION RULES BY LIFT

Antecedent	Consequent	Lift	Conf.
Green Regency Teacup, Regency Cakestand	Pink Regency Teacup	31.03	73.0%
Pink Regency Teacup, Roses Regency Teacup	Green Regency Teacup	30.36	89.3%
Pink Regency Teacup	Green Regency Teacup	28.49	83.8%
Roses Regency Teacup, Cakestand	Pink Regency Teacup	26.98	63.4%
Charlotte Bag Suki Design	Red Retrospot Charlotte Bag	13.47	38.6%

D. Live Recommender Output

The personalized recommendation engine was tested on Customer 17841, classified as High Value, with a current basket containing CHILLI LIGHTS, SUKI SHOULDER BAG, and CHARLOTTE BAG SUKI DESIGN. The top three recommendations were: (1) RED RETROSPOT CHARLOTTE BAG with Lift 13.47 and Confidence 38.6%; (2) STRAWBERRY CHARLOTTE BAG with Lift 12.21 and Confidence 38.0%; (3) WOODLAND CHARLOTTE BAG with Lift 12.21 and Confidence 40.9%. The average lift of 12.63 for this customer exceeds the full-dataset average of 10.57, demonstrating the effectiveness of segment-aware rule generation.

VI. COMPARISON WITH PRIOR WORK

Table VI summarizes how this project extends and improves upon standard Apriori implementations found in recent literature.

TABLE VI. FEATURE COMPARISON WITH STANDARD APRIORI STUDIES

Feature	Standard Apriori Studies	This Work
Customer Segmentation	All customers treated identically	RFM scoring, 3 behavioral tiers
Per-Segment Rules	One global rule set for all	Separate Apriori per segment
Rule Pruning Filter	Support and Confidence only	Support, Confidence, and Conviction
Temporal Analysis	Not performed	Monthly, weekly, and hourly trends
Scalability Measurement	Never reported in literature	Execution time per segment logged
Recommendation Type	Generic for all customers	Personalized per RFM tier
Dataset Coverage	UCI Online Retail, 1 year	UCI Online Retail II, 2 full years

VII. LIMITATIONS AND FUTURE WORK

A. Current Limitations

Product scope is limited to the top 200 items due to memory constraints. A production-scale system would require distributed computing frameworks such as Apache Spark to process the full product catalogue. Temporal analysis in this work is descriptive only; time-aware rules that vary by season or day of week are not implemented in the recommender. The dataset covers 2009 to 2011 and while the methodology is fully transferable to current data, the product catalogue reflects that earlier period.

B. Future Work

Planned extensions include: (i) integration of FP-Growth alongside Apriori for direct performance benchmarking; (ii) construction of seasonal association rules using sliding time windows; (iii) deployment as a REST API for real-time retail recommendation; and (iv) hybrid models combining Apriori rules with collaborative filtering for customers with sparse purchase histories.

VIII. CONCLUSION

This paper presented a Market Basket Analysis system based on the Apriori Algorithm applied to the UCI Online Retail II dataset. The system generated 461 high-quality association rules with an exceptional average lift of 10.57. Four enhancements beyond standard Apriori were implemented and validated: conviction-based rule pruning, RFM-based customer segmentation with per-segment mining, temporal purchase pattern analysis, and scalability measurement. The resulting personalized recommendation engine demonstrates that small-to-medium retailers can deploy interpretable, data-driven recommendations without the complexity of deep learning systems, making it practical for immediate business adoption.

. REFERENCES

- [1] R. Agrawal and R. Srikant, "Fast algorithms for mining association rules," in Proc. 20th Int. Conf. Very Large Data Bases (VLDB), Santiago, Chile, 1994, pp. 487-499.
- [2] D. Chen, S. L. Sain, and K. Guo, "Data mining for the online retail industry: A case study of RFM model-based customer segmentation," J. Database Marketing and Customer Strategy Manage., vol. 19, no. 3, pp. 197-208, 2012.
- [3] M. Hoque et al., "Customer-centric market basket analysis enhancing identification of robust association rules through integration of consumer segmentation," J. Operational and Strategic Analytics, vol. 2, no. 1, 2024.
- [4] R. Vaishampayan et al., "Efficient market basket analysis using association rule mining," in Lecture Notes in Networks and Systems, Springer, 2022.
- [5] UCI Machine Learning Repository, "Online Retail II Dataset," 2019. [Online]. Available: <https://archive.ics.uci.edu/dataset/502/online+retail+ii>
- [6] S. Kannan and S. Immanuel, "An improved Apriori algorithm for market basket analysis," Springer, 2026.